CHE 205, Fall 2019, Problem Set 10

Instructor: Shafigh Mehraeen
Due date: 11:59pm, Fri, 12/06/2019

All problems are from the textbook by V.L. Law.
Submit two spreadsheets, one for each problem, on blackboard.

Problem 1 (5 pts). Exercise 9.1.

Problem 2 (10 pts). Exercise 9.4.

Exercise 9.1: Linear programming.

It is required to produce one pound of an alloy that has at least 30% Pb and at least 30% Zn by mixing a number of available Pb–Zn–Sn alloys. Find the cheapest blend using the following data:

	Analysis (%)			
Available Alloy	**Pb**	**Zn**	**Sn**	**Cost ($/lb)**
1	20	20	60	6.0
2	10	40	50	6.3
3	40	50	10	7.5
4	50	30	20	8.0

Use Solver for this LP problem.

Exercise 9.4: Nonlinear equations.

The calculation of the equilibrium concentration when we have several reactions and components usually results in nonlinear algebraic equations. Consider the following three reactions involving seven components:

$$
\begin{aligned}
&A + B \overset{x_1}{\Leftrightarrow} C + D \qquad && K_1 = \frac{C_C C_D}{C_A C_B} \\
&B + C \overset{x_2}{\Leftrightarrow} E + F \qquad && K_2 = \frac{C_E C_F}{C_B C_C} \\
&A + E \overset{x_3}{\Leftrightarrow} G \qquad && K_3 = \frac{C_G}{C_A C_E}
\end{aligned}
\tag{9.23}
$$

$$K_1 = 1, K_2 = 2, K_3 = 4$$

Here, x_1, x_2, and x_3 (the unknowns) are the *extents of reaction* at equilibrium, and the Cs are molar concentrations. Note that the extent of reaction is a number between 0 and 1. Zero indicates no production of products, while a value of 1 means that the reaction goes to completion (no reactants remain). Given the extents of reaction, the following mass balances can be written:

$$\begin{aligned}
C_A &= C_{A0} - x_1 C_{B0} - x_3 C_{A0} \\
C_B &= C_{B0} - x_1 C_{B0} - x_2 C_{B0} \\
C_C &= C_{C0} + x_1 C_{B0} - x_2 C_{B0} \\
C_D &= C_{D0} + x_1 C_{B0} \\
C_E &= C_{E0} + x_2 C_{B0} - x_3 C_{A0} \\
C_F &= C_{F0} + x_2 C_{B0} \\
C_G &= C_{G0} + x_3 C_{A0}
\end{aligned} \tag{9.24}$$

Initial conditions are $C_{A0} = C_{B0} = 1$; all others are zero. The nonlinear equations 9.23 can be expressed as

$$\begin{aligned}
C_C * C_D - K_1 * C_A * C_B &= 0 \\
C_E * C_F - K_2 * C_B * C_C &= 0 \\
C_G - K_3 * C_A * C_E &= 0
\end{aligned} \tag{9.25}$$

When formulating an initial guess, make note of the following: the optimal values for the variables *must be between 0 and 1.* Use Solver for this problem.